# The Association Between Poverty Status and Self-Reported Depression Rates in the Northeastern United States

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June 2026

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## ABSTRACT

Poverty and mental health are often linked in public discussions, but does living in a state with higher poverty levels actually correspond to higher rates of depression? This study explored that question by examining the relationship between poverty status and self-reported depression rates across the Northeastern United States. Data from the Behavioral Risk Factor Surveillance System (BRFSS) and the U.S. Census Bureau S1701 Poverty Status dataset were combined and analyzed at the state level. Visualizations, descriptive statistics, correlation analysis, and linear regression were used to identify patterns and measure the strength of the relationship between the two variables.

While some states reported both higher poverty and higher depression rates, the overall analysis found no statistically significant relationship between poverty rates and self-reported depression rates in the Northeast. In other words, states with higher poverty levels did not consistently report higher levels of depression. These findings suggest that poverty alone may not explain differences in depression rates across the region and that other factors, such as healthcare access, social support systems, or demographic characteristics, may play an important role. This study highlights the complexity of mental health and demonstrates the importance of examining multiple factors when exploring public health issues.

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## INTRODUCTION

### Background

Poverty falls in the category of economic stability in terms of the social determinants of health. Individuals living in poverty face several disadvantages including limited access to healthcare, the necessities to sustain life, safe living conditions, and more. Financial hardship can create chronic stress and increase the risk of adverse mental health outcomes by limiting access to essential resources and economic stability. Depression can have a profound impact on an individual's daily life, affecting relationships, productivity, physical health, and

overall well-being. Depression represents a significant public health concern due to its effects on individuals, families, and communities. It's important to understand how depression rates can be lowered. For policymakers and public health officials, understanding what outcomes lead to a rise in depression is crucial. This is where studying the correlation between poverty and depression comes in as poverty is a well-known issue in the United States and the world.

### Geographic Focus

According to the United States Census Bureau 2020 Census, the Northeastern United States consists of 9 out of the 50 states. In New England, there is Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont. The remaining three states belong to the Middle Atlantic division: New Jersey, New York, and Pennsylvania.

### Research Question

How is poverty status associated with self-reported depression rates among residents of the Northeastern United States?

### Research Objective

- Examine whether states with higher poverty rates also report higher levels of depression.
- Identify patterns and differences across Northeastern states.

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## METHODOLOGY

### Data Sources

1. Behavioral Risk Factor Surveillance System (BRFSS) - Mental Health Indicators (2022-2024)
- Source: Data.CDC.gov and BRFSS
- Variable: Percentage of adults reporting depression
2. S1701 Poverty Status in the Past 12 Months (2024: ACS 5-Year Estimates Subject Tables)
- Source: United States Census Bureau
- Variable: Percentage of individuals living below the federal poverty line

## Data Preparation

Both datasets were inspected prior to analysis. This consisted of eliminating columns and rows that were unnecessary, missing values were identified and addressed when necessary to ensure data quality, and the formatting of particular values were fixed. Questions asked in the BRFSS study that were focused on some form of depression were isolated from one another. The three questions used in this analysis include: "During the past 30 days, how many days did poor physical or mental health keep you from doing your usual activities, such as self-care, work, or recreation?", "Have you ever been told you have a depressive disorder (including depression, major depression, dysthymia, or minor depression)?", and "Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?". The mean percentage of each state that reported depression was calculated and was compiled into a single analytical dataset. The percentage of each state under the poverty level was extracted from the S1701 dataset, and those values were merged with the values from the means of the BRFSS dataset.

**Figure One: Box Plot of the Percentages of the States' Populations Under the Poverty Level**

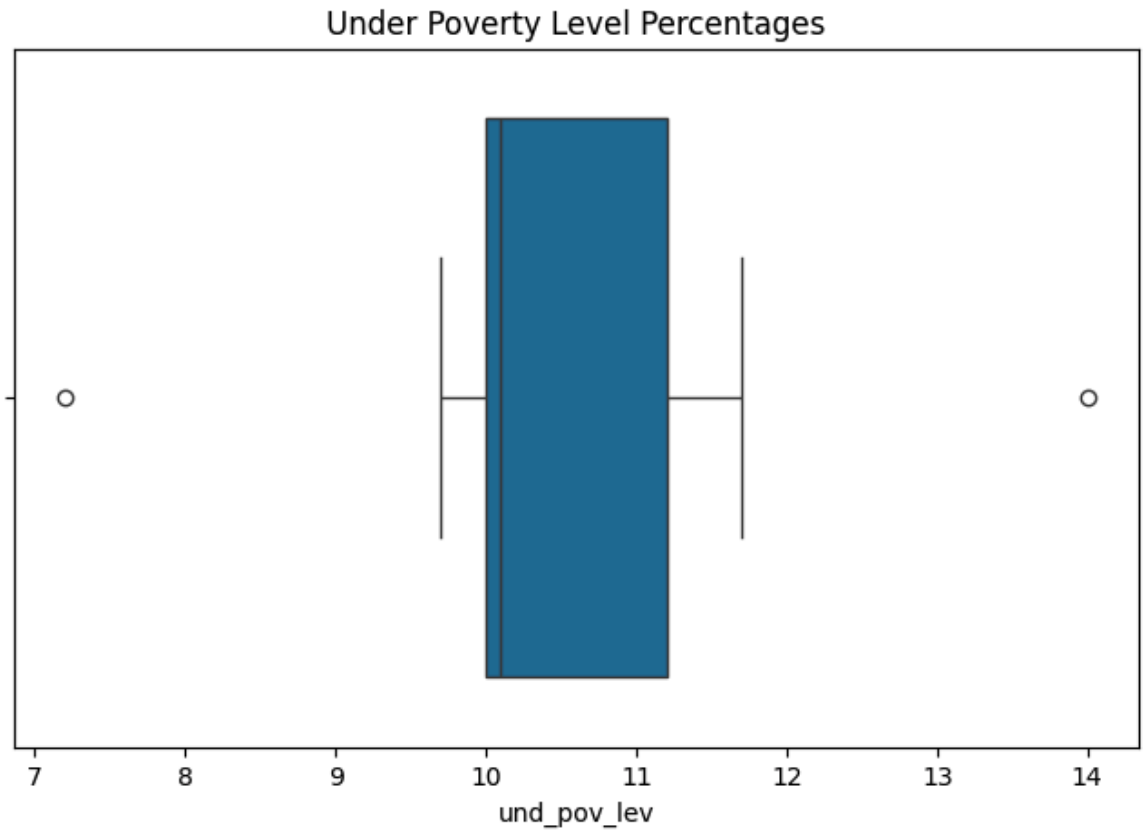
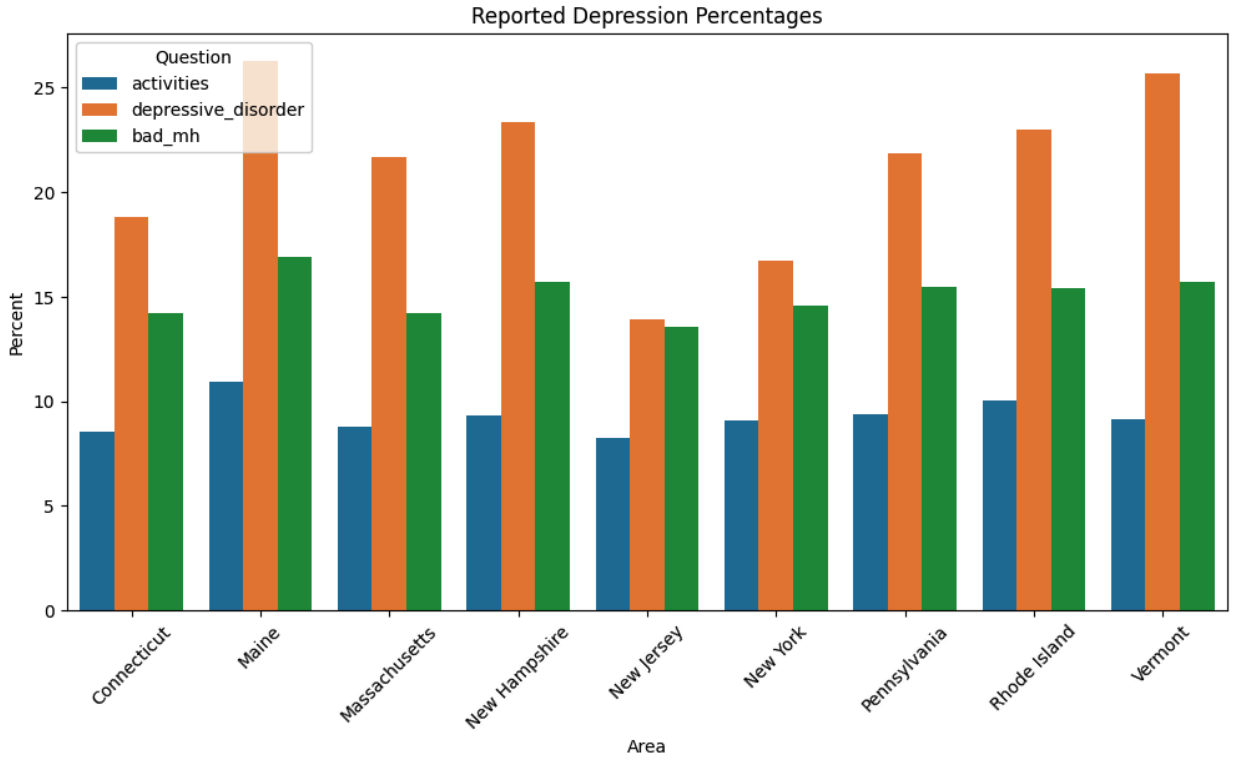



Figure One visualizes that the average percentage of a state's population being under the poverty level is around 10.1%, which is Vermont's poverty level. There are two outliers, one at 7.2% and another at 14%, matching New Hampshire and New York's percentages respectively.

**Figure 2: Bar Graph of Reported Depression Percentages**




The column names correspond to keywords within each question asked in the BRFSS study and the under poverty level percentage. In Figure 2, it can be concluded that Maine has the highest overall percentage of reported depression, and New Jersey has the lowest.

**Figure 3: Merged Datasets - Mean Percentages of Reported Depression and the Percentage Under the Poverty Level**

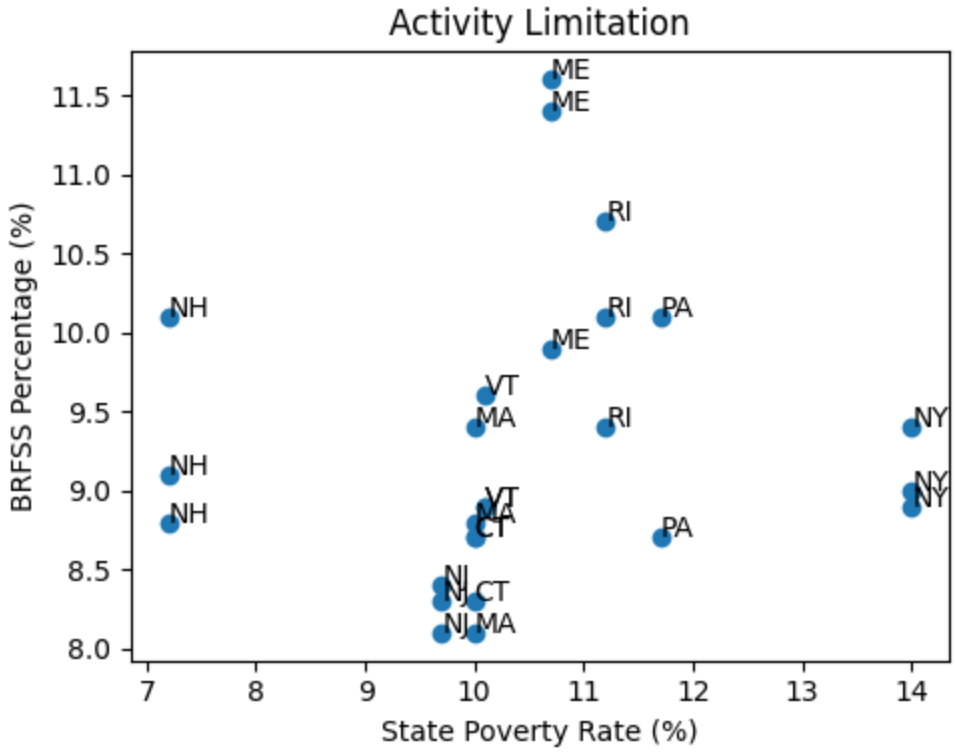
| Area | activities | depressive_disorder | bad_mh | und_pov_lev |
|---|---|---|---|---|
| Connecticut | 8.566667 | 18.800000 | 14.233333 | 10.0 |
| Maine | 10.966667 | 26.266667 | 16.900000 | 10.7 |
| Massachusetts | 8.766667 | 21.700000 | 14.233333 | 10.0 |
| New Hampshire | 9.333333 | 23.366667 | 15.700000 | 7.2 |
| New Jersey | 8.266667 | 13.933333 | 13.566667 | 9.7 |
| New York | 9.100000 | 16.700000 | 14.600000 | 14.0 |
| Pennsylvania | 9.400000 | 21.850000 | 15.450000 | 11.7 |
| Rhode Island | 10.066667 | 22.966667 | 15.400000 | 11.2 |
| Vermont | 9.133333 | 25.666667 | 15.700000 | 10.1 |

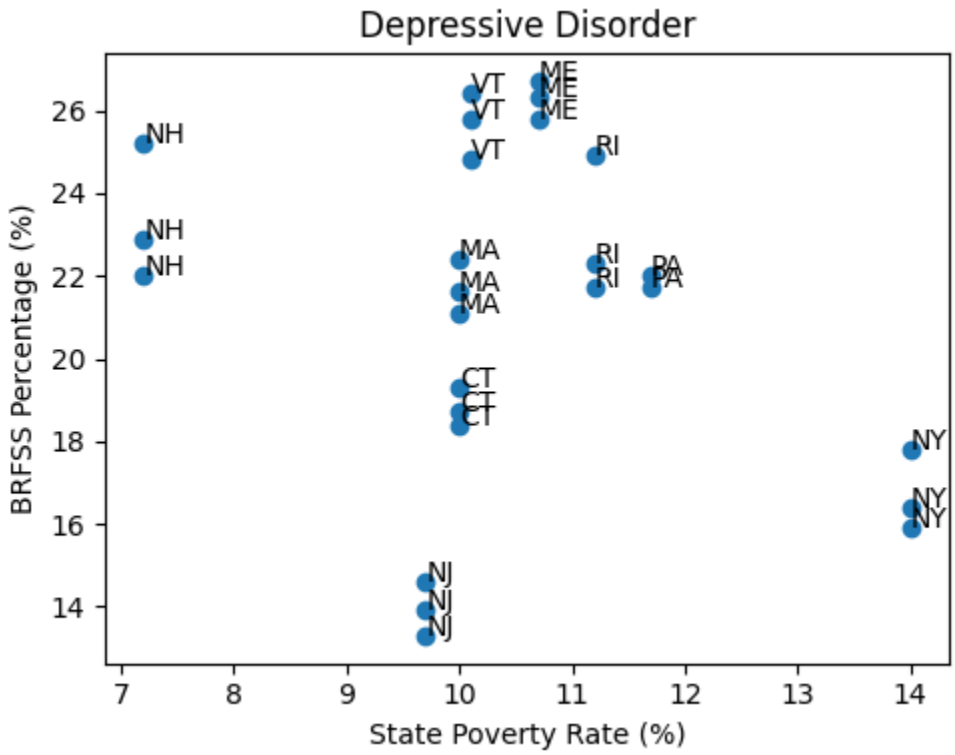
Figure 3 demonstrates that despite Maine having the highest mean percentage of reported depression, it doesn't have the highest population under the poverty level. Likewise, even though New York has the highest population under the poverty level, they don't have the highest mean percentage of reported depression.

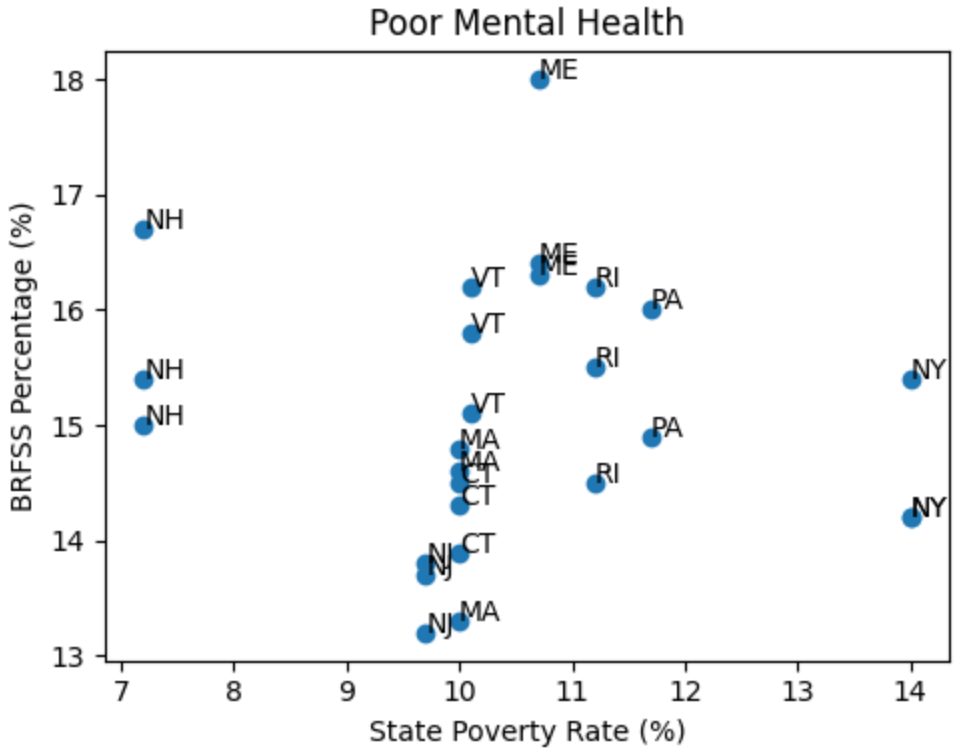
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## KEY FINDINGS

**Figures 4, 5, and 6: Scatterplots Visualizing the Comparison Between Poverty and Depression Rates**

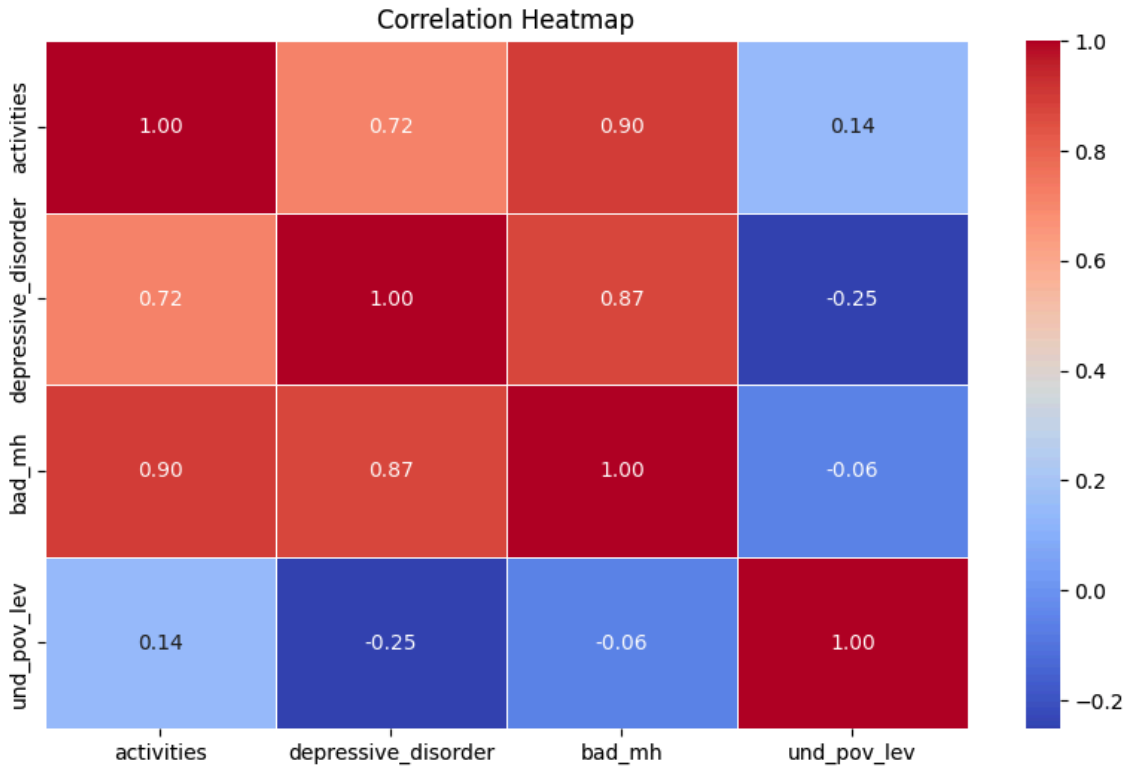

Activity Limitation
BRFSS Percentage (%)
State Poverty Rate (%)
11.5
11.0
10.5
10.0
9.5
9.0
8.5
8.0
7
8
9
10
11
12
13
14
ME
ME
RI
NH
RI
PA
ME
VT
MA
RI
NY
NH
NY
NH
PA
NJ
CT
NJ
MA


Depressive Disorder
BRFSS Percentage (%)
State Poverty Rate (%)
26
24
22
20
18
16
14
7
8
9
10
11
12
13
14
ME
VT
ME
VT
ME
NH
VT
RI
NH
NH
MA
RI
PA
MA
RI
PA
MA
CT
CT
CT
NY
NY
NY
NJ
NJ
NJ




These three figures visualize each state's reported depression rates along with their corresponding poverty levels. It can be inferred that a correlation will not be seen between these variables as reported depression rates don't consistently increase as the poverty rate increases.

**Figure 7: Correlation Matrix/Heatmap Between Variables**




From this visual, the analysis did not find evidence of a statistically significant correlation between poverty and depression. Out of the questions within the BRFSS study, depression's impact on daily activities has the largest positive correlation.

Additionally, the correlations between these variables were calculated individually through Pearson correlation. The results below (r) match the values within the correlation matrix, and the p-values are way above the critical value (.05), meaning it can be concluded that there is no significant correlation between these variables.

```
r1, p1 = stats.pearsonr(df_perc["und_pov_lev"], df_perc["activities"])
print("r1:", r1)
print("p-value1:", p1)
```

```
r1: 0.14389871952711625
p-value1: 0.7118633923482768
```

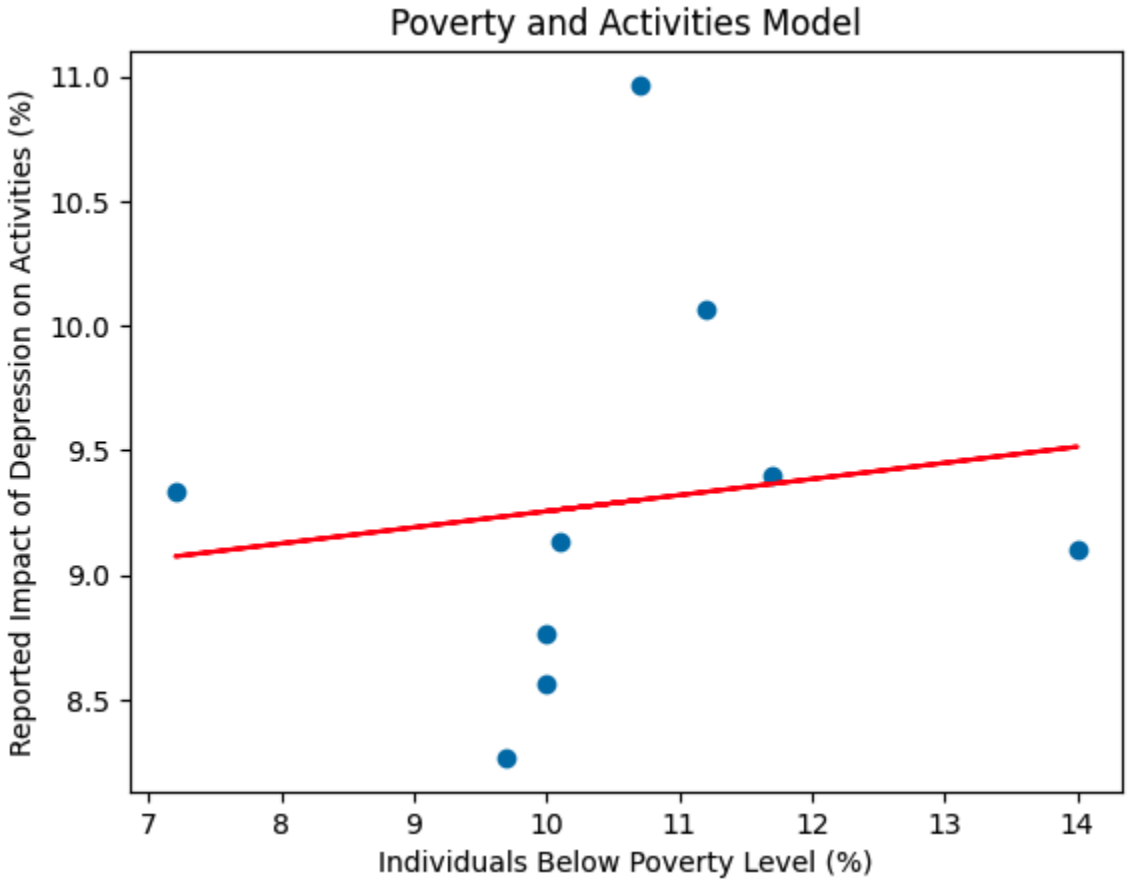
```
r2, p2 = stats.pearsonr(df_perc["und_pov_lev"], df_perc["depressive_disorder"])
print("r2:", r2)
print("p-value2:", p2)
```

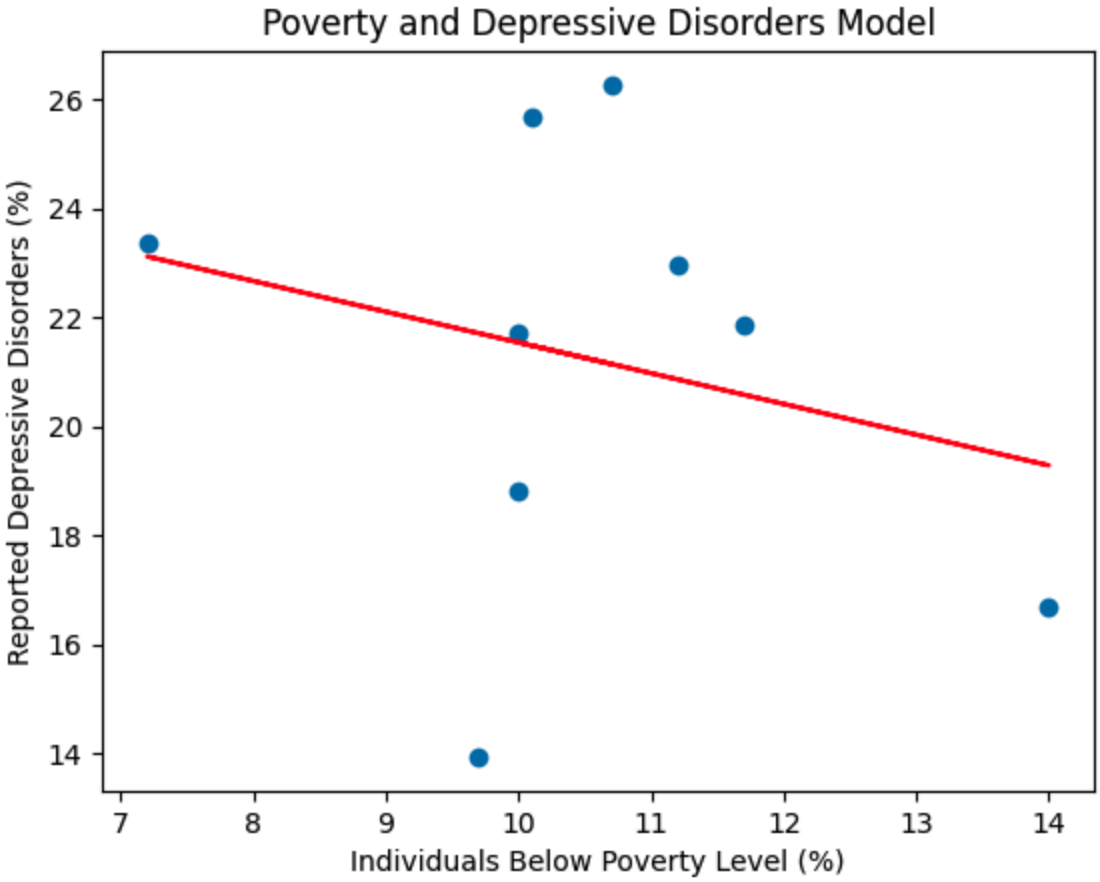
```
r2: -0.25147597512881875
p-value2: 0.5139332682526369
```

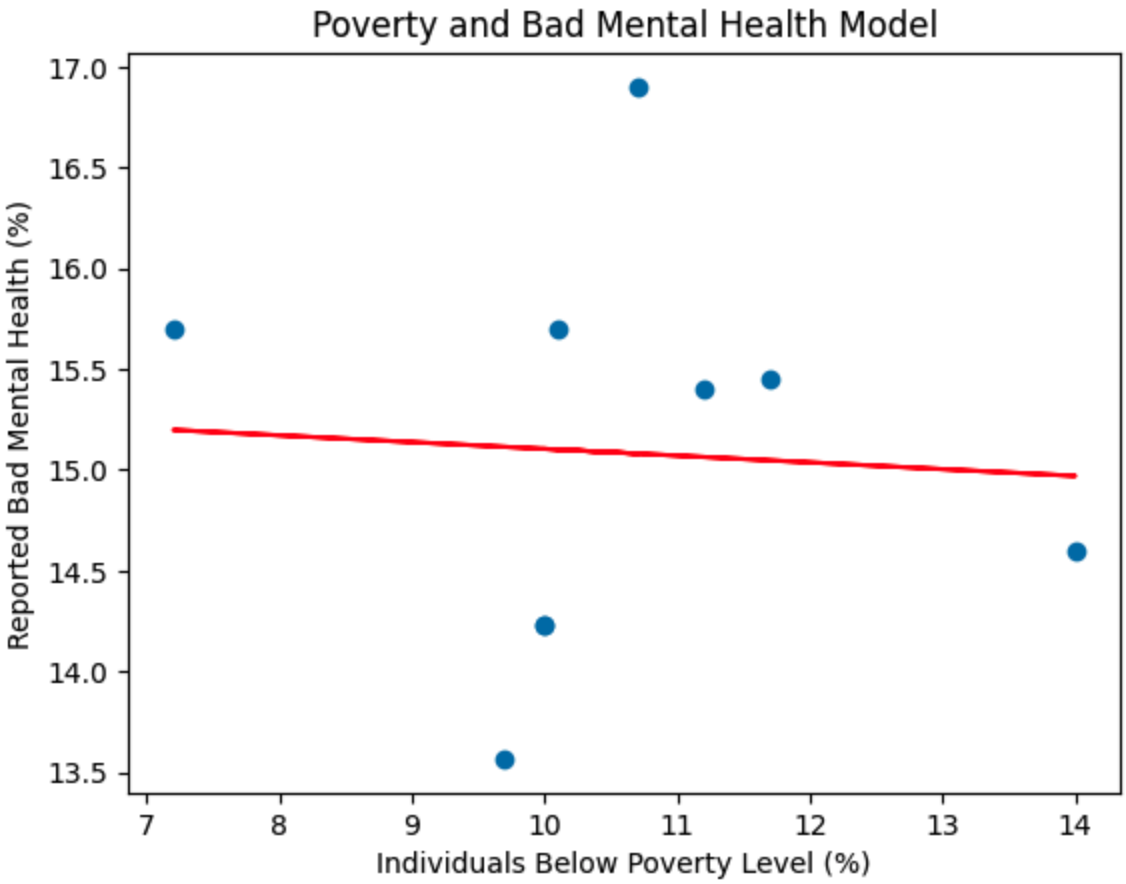
```
r3, p3 = stats.pearsonr(df_perc["und_pov_lev"], df_perc["bad_mh"])
print("r3:", r3)
print("p-value3:", p3)
```

```
r3: -0.0598940577363753
p-value3: 0.8783489929498991
```

**Figures 8, 9, and 10: Linear Regression Models (X = Poverty Level Percentage, y = Reported Depression Percentage)**




Poverty and Depressive Disorders Model
Reported Depressive Disorders (%)
26
24
22
20
18
16
14
7
8
9
10
11
12
13
14
Individuals Below Poverty Level (%)




All three models display points that are heavily scattered, resulting in creating lines that nearly don't change (indicated by the $R^2$ values). As seen from the calculations below, all coefficients are near 0 or below, indicating that an increase in the percent of populational poverty has no major impact on depression levels. There is only one decent sized negative correlation between poverty and reported depressive disorders.

```
# Calculating the coefficient and R^2 of the models
print("Activities - Coefficient:", model1.coef_)
print("Activities - R^2:", model1.score(X, y1))
print("Depressive Disorder - Coefficient:", model2.coef_)
print("Depressive Disorder - R^2:", model2.score(X, y2))
print("Bad MH - Coefficient:", model3.coef_)
print("Bad MH - R^2:", model3.score(X, y3))
```

```
Activities - Coefficient: [[0.06466678]]
Activities - R^2: 0.02070684148154367
Depressive Disorder - Coefficient: [[-0.56414086]]
Depressive Disorder - R^2: 0.0632401660669899
Bad MH - Coefficient: [[-0.03352547]]
Bad MH - R^2: 0.0035872981521286595
```

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## DISCUSSION

From these results, it can be seen that it cannot be suggested that there is a significant correlation between depression and poverty. With this data, the correlation between these variables was explored through the correlation matrix and Pearson correlation. It is seen that the values in the matrix are about 0, and the p-values are well above the critical value. Additionally, the regression models did not provide evidence that increases in poverty levels were associated with increases in reported depression rates.

It is necessary to understand that there could be potential factors that affect these results, both in the world and in the data itself. For instance, while New York has the biggest issue with poverty, citizens receive the most financial assistance through SNAP benefits, Medicaid, housing services, and more. Additionally, mental health outcomes are influenced by numerous social, environmental, and healthcare-related factors. Differences in access to mental health services, insurance coverage, educational attainment, and community support systems may contribute to variation in depression rates that cannot be explained by poverty alone. These ideas can have a major impact on the results as this analysis only looked at poverty itself, not the things around it.

Another important limitation is the relatively small sample size. The analysis focused on a set of states, along with a particular set of years; the statistical power to detect relationships was limited. The BRFSS dataset contained information from 2022-2024, and the S1701 dataset had information from 2019-2024. These characteristics are important to acknowledge because these results cannot be applied to the entire world; with more data and focusing globally, there would be more definitive results on the relationship between these variables.

Although the findings were not statistically significant, they remain valuable because they challenge a commonly held assumption that poverty alone is a primary driver of depression. Null findings contribute to a scientific understanding by identifying areas where additional factors should be investigated.

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## CONCLUSION

This study examined the relationship between poverty status and self-reported depression rates among residents of the Northeastern United States using data from the Behavioral Risk Factor Surveillance System (BRFSS) and the U.S. Census Bureau S1701 Poverty Status dataset. Through exploratory data analysis, correlation testing, and linear regression modeling, the study sought to determine whether states with higher poverty rates also experienced higher rates of depression and other negative mental health outcomes.

The results did not support the existence of a significant relationship between poverty and self-reported depression rates in the Northeastern United States. Correlation analyses revealed weak relationships between poverty rates and all three mental health indicators examined, and none of these relationships were statistically significant. Similarly, the regression models explained very little variation in mental health outcomes, suggesting that poverty alone is not a strong predictor of depression or poor mental health at the state level within this region.

These findings highlight the complexity of mental health and suggest that factors beyond economic status may play a substantial role in shaping mental health outcomes. Variables such as access to healthcare, availability of mental health services, education levels, social support networks, employment opportunities, and demographic characteristics may contribute to differences in reported depression rates across states.

While poverty is often discussed as an important social determinant of health, this analysis demonstrates that its relationship with depression may not be straightforward when examined at the state level. Understanding mental health outcomes requires consideration of multiple interconnected factors rather than a single socioeconomic measure.

Future research could expand this analysis by incorporating additional variables, examining county-level or individual-level data, or exploring other mental health indicators such as anxiety and stress. By taking a broader approach, researchers may gain a more complete understanding of the factors that influence mental health and identify more effective strategies for improving public health outcomes.

Beyond the Northeastern United States, these findings have broader implications for researchers, healthcare professionals, and policymakers around the world. Mental health is a global public health challenge, and poverty is often assumed to be one of its primary causes. However, the results of this study suggest that the

relationship between economic hardship and depression may be more complex than commonly believed. By demonstrating that poverty alone did not significantly predict depression rates in this analysis, the study highlights the importance of examining additional factors such as healthcare access, social support networks, education, community resources, and cultural influences. This broader perspective can help guide future research and inform more effective mental health policies and interventions, both within the United States and internationally. Ultimately, understanding the many factors that contribute to mental health can lead to more targeted solutions and improve quality of life for individuals and communities around the world.